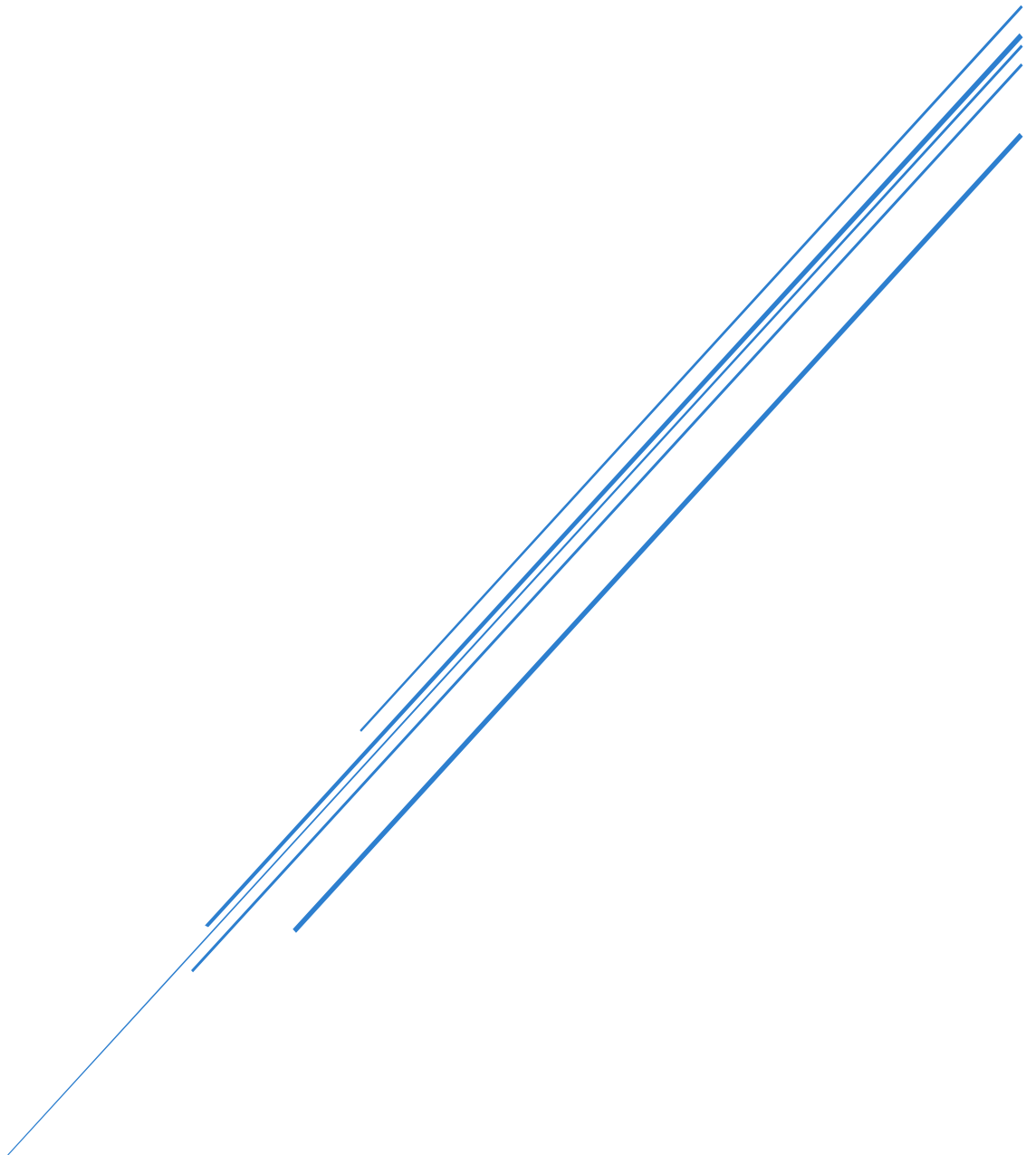


INFORMATION SYSTEMS 2A: ASSIGNMENT 1

Full Name: Kandyce Smit

Student Number: ST10467645



Lecturer: Moses Olaifa
Module Code: INSY6211

Contents

Question 1	2
Question 2	4
Question 3	5
Question 4	7
References	8

Question 1

Q.1.1

1. Title

Beyond the Terminal: Leveraging Systems Analysis and Design to Develop a Cutting-Edge Airport CRM System

2. What is Systems Analysis and Design?

Systems Analysis and Design (SAD) is a systematic method of evaluating, planning, and developing information systems (Turturro, 2023). The analysis phase (what), deals with understanding the requirements and the current processes to determine what the system must accomplish, while the design phase (how), plans the structure, the database and the interfaces that will be required to construct the solution (Indeed, 2025).

For the Airport CRM System, the analysis would be done by conducting interviews with airport staff and passengers to identify the specific requirements related to flight information, GPS navigation and emergency services. Afterwards, the design phase would take those requirements and develop a detailed plan including selecting the kiosk and mobile app platforms, designing the User Interface (UI), and defining the Application Programming Interfaces (API) for GPS, Airline Databases and Medical Response Systems (Myrianthous, 2025).

3. The Importance of Systems Analysis and Design

Properly conducting the Systems Analysis and Design (SAD) for the Airport CRM System is essential for the following reasons:

- Satisfies real users' needs - Gathering Requirements from actual stakeholders ensures the system resolves real passenger and staff issues, rather than perceived issues (Myrianthous, 2025).
- Decreases costs - Identifying potential design problems prior to construction is significantly less expensive than correcting errors after the system has been developed; with the need to include kiosks, a mobile app and several different APIs as part of this system, this is particularly significant (Turturro, 2023).
- Evaluates feasibility - Systems Analysis and Design evaluates the viability of the project in terms of cost, technology, and time prior to full commitment (Indeed, 2025).
- Limits risk - In a high-risk environment such as airports, a failure in a system could result in confusion among passengers or delayed emergency responses; Systems Analysis and Design identify these risks prior to their occurrence (Myrianthous, 2025).
- Ensures alignment of stakeholders - Systems Analysis and Design develops a common understanding of scope among developers, airport management, and end-users prior to beginning the development of the system (Turturro, 2023).

4. Activities of System Analysis

During the Systems Analysis Phase of the Airport CRM System, the following activities will take place:

- Requirements Gathering - Interviews and/or observation studies will be done with the following groups: airport employees, IT personnel, and passengers to identify what the system needs to perform and include, such as show flight information, display GPS navigation and emergency medical services (TutorialsPoint, 2026).
- Feasibility Study - A feasibility study will be completed prior to allocating financial and technical resources in order to establish whether the proposed system can be implemented at an acceptable cost; whether the technical requirements of the system are feasible; and whether it can be delivered on time (TutorialsPoint, 2026).
- Problem Definition - The exact problems that the system will need to resolve will be clearly defined, for example, if there are issues related to a passenger's ability to navigate the airport and/or gain access to emergency assistance (Myrianthous, 2025).
- Use Cases - All of the ways in which users interact with the system will be documented (for example, check flight status, find your way to a gate, view airport amenities, request assistance from a medical service) (Myrianthous, 2025).
- Domain Modelling - Key entities that exist within the system will be identified, including passengers, flights, boarding gates, amenities and emergency services and describe how these entities interrelate (TutorialsPoint, 2026).

5. Activities of System Design

During the Systems Design Phase of the Airport CRM System, the following activities will take place:

- Architectural Design - The Airport CRM System architecture will be defined in detail. The design will include a web-based interface that users will interact with through an airport kiosk, as well as a mobile application available for download from the Apple App Store and Google Play Store (TutorialsPoint, 2026).
- Database Design - The database design will be developed to allow for the collection and management of data related to flight schedules, gate assignments, and amenity locations. Additionally, the database design will integrate the Airport CRM System with current airline databases (TutorialsPoint, 2026).

- Interface Design (UI) - User-friendly interfaces will be designed for both the kiosk and the mobile application. These designs will enable passengers to quickly locate information about flights and the airport, navigate the airport, and quickly contact emergency services (Turturro, 2023).
- API and Integration Design - The APIs used to connect the Airport CRM System to external services such as real time GPS navigation, live flight databases, and the airport's medical emergency response system will be identified (TutorialsPoint, 2026).
- Security Design - Security processes will be identified to protect any passenger data collected by the Airport CRM System. In addition to protecting the confidentiality of passenger data, security processes will be put into place to control access to the passenger data and encrypt it so that unauthorised parties cannot access it (TutorialsPoint, 2026).

6. Iterative approach and its motive for the Airport CRM System

The iterative approach divides development into many smaller repeated cycles of developing the plan, creating the build, and testing the product as opposed to completing the overall system in one single phase (Awati, 2023). An iterative approach to development is different from a waterfall approach to development where all the planning, development and testing occur sequentially, so any issues are typically found and corrected late in the development cycle, wasting time and money (Awati, 2023).

An iterative approach is recommended for the Airport CRM System due to the following reasons:

- It accommodates changing requirements - As the development of the system proceeds, the needs of the airport may evolve and therefore the requirements of the Airport CRM System will change during the course of the project, for example management may want to include additional amenities or make changes to emergency procedures; the iterative methodology provides the opportunity to incorporate these types of changes at any point in the project without delaying the completion of the project (Awati, 2023).
- It enables priority-based feature releases - Critical components such as flight information and emergency medical services can be developed and delivered first, followed by non-essential features such as GPS navigation and amenity listing in future iterations (Awati, 2023).
- It provides opportunities for stakeholders to provide feedback on each iteration of the product - Airport staff and passengers can review and test each iteration of the product prior to the start of the next cycle, providing an opportunity for improving or correcting the product before the next cycle of the project begins (Awati, 2023).
- It reduces financial risks - Issues that arise with the product or design are identified and addressed immediately within each cycle of the product, compared to waiting until the end of the product's development and then attempting to correct them, therefore greatly reducing the costs associated with resolving errors (Awati, 2023).

The Airport CRM System should utilise the Agile Framework approach for structuring the development process, as the Agile Framework utilises short sprints (typically 2-4 weeks) to deliver working products and continuous improvement through the duration of the project (Awati, 2023).

7. Conclusion

This report demonstrates the importance of a systematic and structured Systems Analysis and Design method to develop a functional Airport CRM System. Systems Analysis is a process of identifying the system's necessities; while Systems Design is a process of determining how the system should be developed, including designing the platform architecture, the data storage, the UI, and integrating APIs, to create a solid base for a good information system. (Turturro, 2023; Indeed, 2025).

The benefits of using a correct and complete SAD process for this project are financial viability, meeting the actual demands of passengers and employees, reduced development risks and alignment of stakeholders throughout the project cycle (Myrianthous, 2025). Systems Analysis will allow the system to be completely understood by ensuring the system's specifications are accurately defined before any development occurs. The Systems Design Process will provide the tools to convert these specifications into a workable, reliable and usable system (TutorialsPoint, 2026).

In addition to the use of SAD methods, the adoption of an Agile iterative approach, where sprints are typically 2-4 weeks long, will enable the system to evolve with changes in the airport environment, enable the implementation of the most important features, such as flight status and emergency medical service, and enable the provision of continued input from stakeholders to improve the system continuously during the development phase (Awati, 2023).

My team and I are confident that we have the necessary skills, approaches and knowledge of the principles of both systems analysis and design to successfully implement a world-class CRM system for this airport. Our experience of aligning the software development process with an organisation's overall strategy and the commitment to providing continuous input from stakeholders on the development of the system make us the right partner to bring your vision to reality. We would appreciate the chance to serve this airport and their passengers.

Question 2

Q.2.1

The key stakeholders in this case will be those who are interested in, impacted by, or both interested in and impacted by, the success of this Airport CRM System (Long, 2017). These may be individuals or groups within (Internal) or outside (External) of the organisation. The following five Stakeholders were identified for the Airport CRM System; their interests and contributions toward requirements gathering are also given below:

- Airport Manager (Internal - Executive Stakeholder): The Airport Manager sets the strategic direction for the entire Airport; outlines the overall objectives, strategic priorities, budgetary limits for the system; ensures the Development Team develops the exact system airport management requires (Long, 2017).
- Passengers/Tourists (External - Operational Stakeholder): Passengers are the primary users of the system. They will indicate what functionality is needed, such as flight information, directions, etc., and they will evaluate each iteration of the system to provide input regarding any usability enhancements (Long, 2017).
- IT Department (Internal - Operational Stakeholder): The IT Department provides the Development Team with insight into the technical capabilities of the airport's current technology infrastructure, the current systems being utilised at the airport, and any technical limitations; all of these elements will impact decisions related to platform compatibility, database integration, and security design (Long, 2017).
- Emergency Medical Services (External - Operational Stakeholder): Emergency Medical Services provides the Development Team with an understanding of how their emergency response systems function today, so that the Design Team can create the emergency connection feature in a manner that will allow medical professionals to respond as efficiently and effectively as possible (Long, 2017).
- Airport Receptionist/Check-in Staff (Internal - Operational Stakeholder): The Airport Receptionist/Check-in staff provide the Development Team with an understanding of the most frequent inquiries received from passengers, so that the Development Team can determine which features and data are most important to include in the system (Long, 2017).

Question 3

Q.3.1

Event Decomposition is a method of identifying Use Cases by examining the Business Events which result in a System Response (Learn Information System, 2016). Each Use Case represents how the user interacts with the system to meet an objective, and it describes how the System responds to each Business Event (Brush, 2022). Business Events are typically categorised in one of the following ways: External (Triggered by person or other systems), Temporal (Triggered by time), or State (Triggered by change in condition). (Satzinger, Jackson and Burd, 2015).

The following four business events and their related Use Cases were identified for the Airport CRM System:

1. Event (External) - A passenger wants to check the status of their flight
Use Case - View Flight Information
2. Event (External) - A passenger needs to locate airport amenities such as bathrooms or restaurants
Use Case - Browse Airport Amenities
3. Event (External + State) - A medical emergency occurs at the airport
Use Case - Request Emergency Medical Assistance
4. Event (State) - A flight is cancelled or delayed, and passengers need to be notified
Use Case - Receive Flight Status Notification

Q.3.2

The following use case description was developed using the fully developed use case description template given by Satzinger, Jackson and Burd (2015):

Use Case Name:	Request Emergency Medical Assistance
Scenario:	A passenger has a seizure and requires immediate medical attention
Triggering event:	An individual suffers a medical emergency at the airport, and a bystander activates the Emergency Medical Assistance (EMAS) option on the mobile application or kiosk.
Brief description:	Bystander sends an emergency alert via the Airport CRM system that includes the bystander's location and the nearest gate. EMS will then receive the alert along with the location information and send help to the scene.
Actors:	Bystander, EMS Team.
Related use cases:	Could be supported by the "Browse Airport Amenities" use case to find the nearest gate or landmark so the EMS can locate where the emergency occurred.
Stakeholders:	Patient experiencing the medical emergency, Airport Management, Airport Security Staff
Preconditions:	Airport CRM mobile application or kiosk must be activated and working correctly Emergency Assistance function within Airport CRM must be connected to EMS response system GPS/Location Tracking must be enabled and functioning EMS team must be available and able to receive alerts.

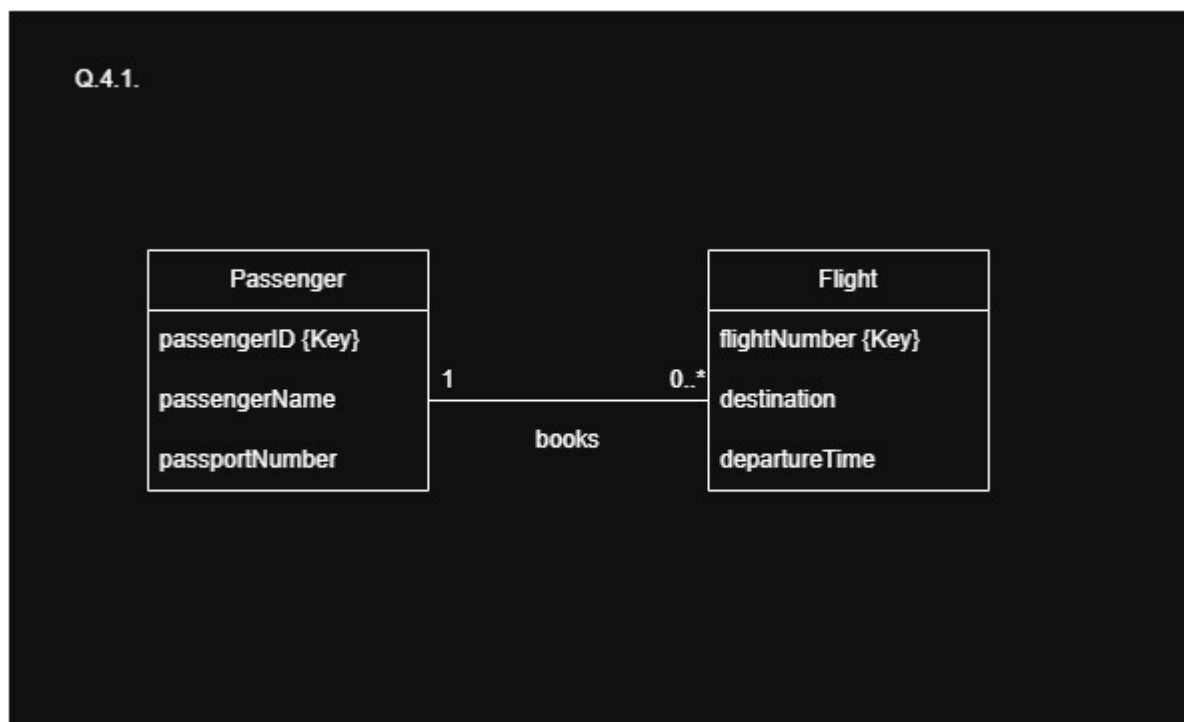
Postconditions:	Alert was successfully sent and received by EMS personnel. Location of Patient and Nearest Gate were provided to EMS personnel. Airport Management has been notified of the emergency as well as its resolution. Emergency Event has been recorded in the Airport CRM System for documentation.	
Flow of activities:	Actor	System
	1. Bystander notices passenger having a seizure 2. Bystander opens the Airport CRM app and selects the emergency assistance feature 3. Bystander confirms the emergency and submits the alert 4. EMS receives the alert and dispatches to the location 5. EMS Arrives at Scene and Provides First Aid to Individual.	2.1 System detects and records the bystander's location and nearest gate 3.1 System sends emergency alert with location details to EMS 3.2 System updates app status to "EMS On-Route" 4.1 System tracks and updates EMS arrival status in real time 5.1 System logs the full emergency event and notifies airport management
Exception conditions	2.1 GPS is inoperable or unable to detect the emergency location. 3.1 Network connectivity issues prevent the alert from being sent to EMS 4.1 The Airport CRM mobile application or kiosk crashes	

Question 4

Q.4.1

Domain Model Class Diagram for Question 4:

The following Domain Model Class Diagram was developed using the domain model class diagram notation given by Satzinger, Jackson and Burd (2015) and drawn using draw.io :



References

Awati, R., 2023. *What is iterative development?*. [Online]

Available at: <https://www.techtarget.com/searchsoftwarequality/definition/iterative-development>

[Accessed 09 March 2026].

Brush, K., 2022. *What is a use case?*. [Online]

Available at: <https://www.techtarget.com/searchsoftwarequality/definition/use-case>

[Accessed 11 March 2026].

Indeed, 2025. *What Is System Analysis and Design? (Plus Benefits)*. [Online]

Available at: <https://www.indeed.com/career-advice/career-development/what-is-system-analysis-and-design>

[Accessed 05 March 2026].

Learn Information System, 2016. *Use Cases and Event Decomposition*. [Online]

Available at: <https://learn-informationsystem.blogspot.com/2016/05/use-cases-and-event-decomposition.html>

[Accessed 11 March 2026].

Long, J., 2017. *Importance of Airport Stakeholder Outreach*. [Online]

Available at: <https://www.aviationpros.com/airport-business/airport-infrastructure-operations/blog/12388063/importance-of-airport-stakeholder-outreach>

[Accessed 11 March 2026].

Myrianthous, 2025. *What Is a Systems Analysis?*. [Online]

Available at: <https://builtin.com/articles/systems-analysis>

[Accessed 05 March 2026].

Satzinger, J., Jackson, R. & Burd, S., 2012. *Systems Analysis and Design in a Changing World - Chapter 3: Use Cases (Slides)*. [Online]

Available at: <https://feng.stafpu.bu.edu.eg/Electrical%20Engineering/3064/crs-10330/Files/chapter%203-1%20Use%20Cases.pdf>

[Accessed 11 March 2026].

Satzinger, J. W., Jackson, R. & Burd, S., 2015. *Systems Analysis and Design in a Changing World*. 7th ed. Boston: Cengage Learning.

Turturro, 2023. *Systems Analysis and Design: Exploring Modern Systems*. [Online]

Available at: <https://www.door3.com/blog/system-analysis-and-design>

[Accessed 05 March 2026].

TutorialsPoint, 2026. *System Analysis & Design - System Design*. [Online]

Available at: https://www.tutorialspoint.com/system_analysis_and_design/system_design.htm

[Accessed 06 March 2026].

TutorialsPoint, 2026. *System Analysis & Design - System Planning*. [Online]

Available at:

https://www.tutorialspoint.com/system_analysis_and_design/system_analysis_and_design_planning.htm

[Accessed 06 March 2026].

